# 10.6 Circumference and arc length

EASY

**Learning intention** We are learning to find circumferences, arc lengths and perimeters of curved shapes.

**Success criteria**
- ☐ I can use C = πd or C = 2πr to find a circumference
- ☐ I can find an arc length: θ ÷ 360 × 2πr
- ☐ I can find the perimeter of sectors and composite shapes

## NOTES — Key ideas for this lesson

TEACHER COPY

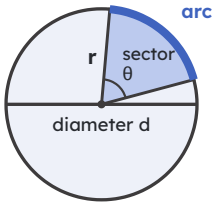
### KEY TERMS

**circumference (C):** the distance around a circle
**radius (r):** from the centre to the edge
**diameter (d):** across through the centre; d = 2r
**arc:** part of the circumference
**sector:** a slice of a circle between two radii
**θ:** the angle at the centre of a sector
**perimeter:** the distance around the outside of a shape

### FORMULAS

**C = πd = 2πr**
**Arc length = $\frac{\theta}{360}$ × 2πr**
**Semicircle: P = ½πd + d**
**Quadrant: P = ¼ × 2πr + 2r**
**Sector: P = arc + 2r**

**STEPS** 1 r = d ÷ 2 2 write the formula 3 substitute (π key) 4 round to 1 d.p. + unit



circumference

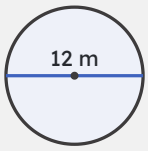
## WE DO — With your teacher: copy what they write on the board.

**Example 1** Find each circumference or perimeter, correct to one decimal place.

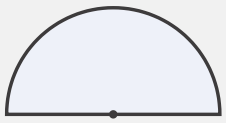
**a** Circumference



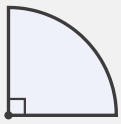

**b** Circumference




**c** Perimeter




**d** Perimeter




## YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

**REMEMBER** d = 2r · r = d ÷ 2 · C = πd · Use the π key, then round to one decimal place.

### ROUND 1 — Questions 1–4: find the diameter or the radius.

**1** r = 4 cm. Find d.
Formula: d = 2r
Substitute:
Answer:

**2** d = 10 m. Find r.
Formula: r = d ÷ 2
Substitute:
Answer:

**3** r = 6.5 cm. Find d.
Formula: d = 2r
Substitute:
Answer:

**4** d = 9 m. Find r.
Formula: r = d ÷ 2
Substitute:
Answer:

### ROUND 2 — Questions 5–8: the diameter is given. Find the circumference.

**5** d = 6 cm
Formula: C = πd
Substitute:
Answer:

**6** d = 10 m
Formula: C = πd
Substitute:
Answer:

**7** d = 2.5 m
Formula: C = πd
Substitute:
Answer:

**8** d = 14 cm
Formula: C = πd
Substitute:
Answer:

**YOU DO** On your own: keep going, one round at a time. Show your working. 15 QUESTIONS

**REMEMBER** C = 2πr = πd · Semicircle P = ½πd + d · Quadrant P = ¼ × 2πr + 2r · Find d first when r is given.

## ROUND 3 Questions 9–11: the radius is given. Find the circumference.

**9** r = 2 cm

Formula C = 2πr
Substitute
Answer

**10** r = 8 m

Formula C = 2πr
Substitute
Answer

**11** r = 1.5 m

Formula C = 2πr
Substitute
Answer

## ROUND 4 Questions 12–14: is it r or d? Write the right formula, then find the circumference.

**12** r = 4 cm

Formula
Substitute
Answer

**13** d = 9 m

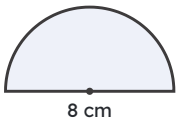
Formula
Substitute
Answer

**14** r = 12 mm

Formula
Substitute
Answer

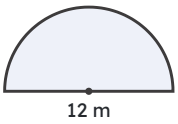
## ROUND 5 Questions 15–17: find the perimeter of each semicircle.

**15**



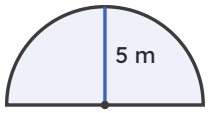

Formula P = ½πd + d
Substitute
Answer

**16**




Formula P = ½πd + d
Substitute
Answer

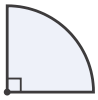
**17**




Formula P = ½πd + d
Substitute
Answer

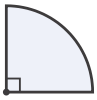
## ROUND 6 Questions 18–20: find the perimeter of each quadrant.

**18**



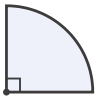

Formula P = ¼ × 2πr + 2r
Substitute
Answer

**19**




Formula P = ¼ × 2πr + 2r
Substitute
Answer

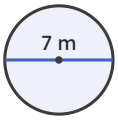
**20**




Formula P = ¼ × 2πr + 2r
Substitute
Answer

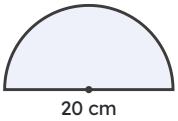
## ROUND 7 Questions 21–23: circle, semicircle or quadrant? Write the formula, then find the perimeter.

**21**



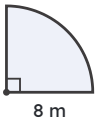

Formula
Substitute
Answer

**22**




Formula
Substitute
Answer

**23**




Formula
Substitute
Answer

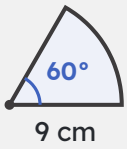
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MEDIUM

**WE DO** With your teacher: copy what they write on the board.

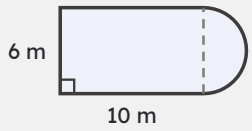
**Example 2** **Give each answer correct to one decimal place.**

**a** Arc length




**b** Perimeter of the sector in part a

**c** Perimeter




**d** The radius of a circle with C = 50 cm

**YOU DO** On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

**REMEMBER** Arc = θ ÷ 360 × 2πr (θ = angle at the centre) · Sector P = arc + 2r · Composite: add the outside edges only

**1** Find the arc length of a 120° sector with r = 3 m.

Formula _____
Substitute _____
Answer _____

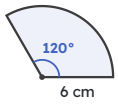
**2** A kite has diagonals of 6 m and 5 m. Find its area. 10.5

Formula _____
Substitute _____
Answer _____

**3** Find the arc length of a 45° sector with r = 10 cm.

Formula _____
Substitute _____
Answer _____

**4** Perimeter



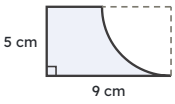

Formula _____
Substitute _____
Answer _____

**5** Perimeter




Formula _____
Substitute _____
Answer _____

**6** Perimeter. Cut-out: r = 5 cm




Formula _____
Substitute _____
Answer _____

**7** Write the limits of accuracy of 4.5 m. 10.2

_____

**8** A circle has a circumference of 18.85 m. Find its radius.

Formula _____
Substitute _____
Answer _____

**9** Percentage error of 20 cm, to the nearest cm? 10.3

Formula _____
Substitute _____
Answer _____

**WORDED PROBLEMS** Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

**1** A round garden bed has a diameter of 3.2 m. Edging costs $12 per metre. Find the cost to edge the bed, to the nearest dollar.

Answer _____

**2** A bike wheel has a radius of 33 cm. How many full turns does it make to travel 1 km?

Answer _____

**3** A running track is a rectangle 84 m long with a semicircle on each end. Each semicircle has a diameter of 64 m. Find one lap, to the nearest metre.

Answer _____